

the pre-sensitised PCB board to a size slightly bigger than the boundary of the artwork. Take a 20x25cm² (8x10 inches²) rectangular piece of Perspex (or plastic or Bakelite) sheet and a transparent piece of glass of the same size around 4mm thick. Using double-sided tape, stick the laminate side of the board onto the Perspex sheet.

Next, you need to place the artwork on top of the PCB side covered with the protective opaque sticker. The correct way to do this is to have the side (surface) of the transparency on which the artwork is printed

(ink deposited by the laser printer) to face and touch the PCB. This ensures that the orientation of the artwork is correct and the ink on the artwork touches the pre-sensitised side of the artwork, leaving no space for light to 'ooze' or 'creep' in between the ink on the transparency and the laminate. Align the artwork to the PCB.

Use a regular (one-sided) piece of tape to stick one edge of the artwork to the Perspex sheet, forming a kind of hinge. So, when the opaque sticker is eventually removed, the artwork is positioned

correctly on top of the sensitised surface of the PCB.

Now you are ready to expose the PCB. The apparatus I used was an 11-watt CFL bulb fitted in an ordinary reading lamp (see Fig. 23). Natural light can expose (and harm) the pre-sensitised board, so perform this step either at night when there is no natural light or in a dark room setting during the day. In fact, just as a

side bar, even sunlight can be used to expose the PCB, but this method is contingent on getting the settings right for the amount of sunlight and the duration of exposure. You can use a small red-coloured bulb in the working area without harming the PCB. Now arrange the apparatus as shown in Fig. 23, ensuring that the CFL is around 5cm above the glass.

It will be nice if there is someone to help you hereafter. Keep a timer next to you; I used the timer utility on my smart phone. Before exposing the PCB, switch on the CFL bulb for around 30 minutes while keeping your sandwich away from it in a dark area. Next, in a dark room setting, slowly remove the protective opaque sticker, ensuring that the PCB does not move from its place.

Ensure that the artwork is aligned with the PCB. Place the glass piece on top of the artwork. Then either use rubber bands or the commonly available paper binder clip to hold the sandwich together. This will ensure that the glass presses the artwork flush against the PCB which has been fastened to the Perspex sheet. Just to reiterate, the final sandwich will have the glass on the top, the artwork transparency below it touching the PCB's photo-sensitised side with the laminate's

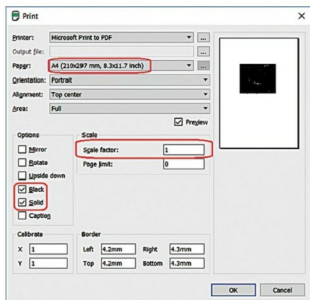


Fig. 20: Settings for creating a PDF file

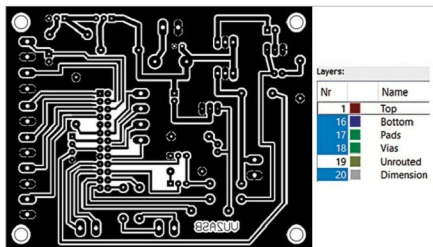


Fig. 21: The final artwork (left) and (right) selection of only the top, bottom, pads, vias and dimension layers before generating the artwork



Fig. 22: Inspecting and aligning your artwork